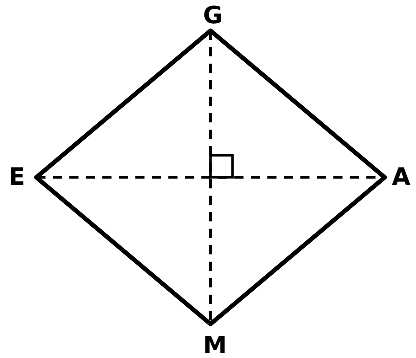
The Rhombus

A rhombus has four equal sides. It is a parallelogram with a special bonus: its diagonals cross at right angles and bisect the corner angles.

Properties of a rhombus



A rhombus GAME; the diagonals meet at 90°.

1. **All four sides are equal.**
2. It is a **parallelogram**, so opposite angles are equal and adjacent angles add to 180°.
3. The **diagonals bisect each other at right angles (90°)**.
4. The **diagonals bisect the corner angles**.

Handy right triangle: Because the diagonals are perpendicular and bisect each other, each half-diagonal forms a right triangle. So side = $\sqrt{[(\frac{1}{2}d_1)^2 + (\frac{1}{2}d_2)^2]}$.

Family link: Square ⊂ Rhombus ⊂ Parallelogram. A square is a rhombus with right angles.

Where marks slip away

Trap 1 — Rhombus diagonals are not equal. They bisect each other at 90°, but they are usually **different lengths** (equal only in a square).

Trap 2 — Using the full diagonal in the side formula. Use **half** of each diagonal in $\sqrt{[(\frac{1}{2}d_1)^2 + (\frac{1}{2}d_2)^2]}$.

Trap 3 — Forgetting it is a parallelogram. All parallelogram angle rules still apply (opposite equal, adjacent supplementary).

Trap 4 — Right angle in a rhombus. A rhombus with one 90° angle is actually a **square**.

Slip 5 — Name the right-angle property when diagonals are involved to earn the mark.

Model answers — write them like this

Example A. The diagonals of a rhombus are 6 cm and 8 cm. Find the length of each side.

Step 1: Half-diagonals are 3 cm and 4 cm, meeting at a right angle.

Step 2: side = $\sqrt{(3^2 + 4^2)} = \sqrt{(9 + 16)} = \sqrt{25}$.

Step 3: = 5 cm.

∴ Each side = 5 cm

Example B. In a rhombus, one angle is 50°. Find the other three angles.

Step 1: A rhombus is a parallelogram: opposite angles are equal, so another angle is 50°.

Step 2: Adjacent angles are supplementary: 180° − 50° = 130°.

∴ 50°, 130°, 50°, 130°

Example C. A diagonal of a rhombus is equal in length to a side. Find the angles of the rhombus.

Step 1: The diagonal splits the rhombus into two triangles. With diagonal = side, each triangle has all three sides equal (the two sides of the rhombus and the diagonal), so it is equilateral.

Step 2: Each angle of an equilateral triangle is 60°, so one angle of the rhombus is 60°.

Step 3: Adjacent angle = 180° − 60° = 120°.

∴ 60°, 120°, 60°, 120°

Next: try the worksheet, then check the answer key.